

Technical 2: Android Activity Lifecycle

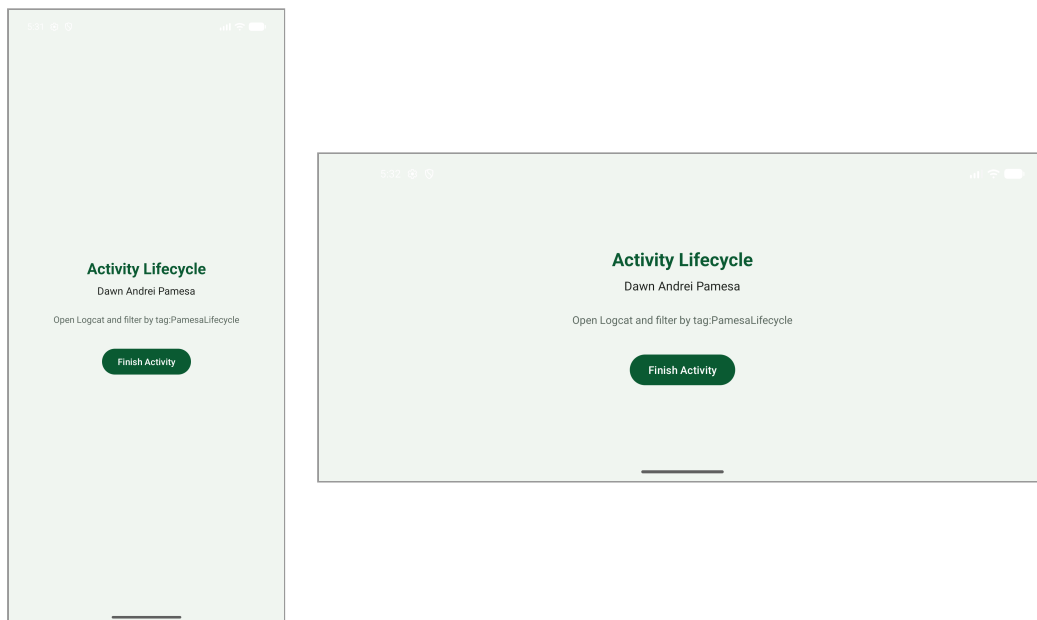
Lifecycle Observation Report

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Application Overview

The app has one Activity, MainActivity, that overrides all seven lifecycle callbacks: onCreate(), onStart(), onResume(), onPause(), onStop(), onRestart() and onDestroy(). Each callback calls Log.d() with the tag PamesaLifecycle and the message Dawn Andrei Pamesa is calling <method>(), so all messages can be filtered in Logcat with tag:PamesaLifecycle. A Finish Activity button calls finish() so that onDestroy() can be triggered on purpose. The app was tested on the Pixel 10 Pro emulator.

Figure 1. The app in portrait and landscape.



Observation Table

#	Action Performed	Callback Sequence Observed	Explanation
1	Launch the app	onCreate() → onStart() → onResume()	The system creates the Activity, makes it visible, then brings it to the foreground so the user can interact with it.
2	Press Home	onPause() → onStop()	The Activity loses focus first (onPause), then is fully hidden by the home screen (onStop). It stays in memory, so it is not destroyed.
3	Reopen the app	onRestart() → onStart() → onResume()	The Activity was only stopped, so it comes back through onRestart() instead of onCreate(), then

			becomes visible and interactive again.
4	Switch to another app (Settings)	onPause() → onStop()	Settings covers the whole screen, so the Activity is paused and then stopped, the same as pressing Home.
5	Switch back to the app	onRestart() → onStart() → onResume()	The stopped Activity is brought back to the front, so it restarts, starts and resumes without being created again.
6	Rotate the device to landscape	onPause() → onStop() → onDestroy() → onCreate() → onStart() → onResume()	Rotation is a configuration change. By default Android destroys the Activity and creates a new one so the layout reloads for the new orientation. Rotating back to portrait produced the same six callbacks.
7	Press Back	onPause() → onStop() → onDestroy()	Back tells the system the user is done with this screen, so the Activity is paused, stopped and destroyed. The process stayed alive (same PID 6229).
8	Launch the app again	onCreate() → onStart() → onResume()	The previous Activity was destroyed, so a brand new one starts with onCreate() instead of onRestart().
9	Tap the Finish Activity button	onPause() → onStop() → onDestroy()	finish() ends the Activity on purpose, so onDestroy() is guaranteed to run here, unlike when the system kills the process.

Logcat Evidence

Each figure shows the Logcat messages produced by one action, filtered by the PamesaLifecycle tag. All seven callbacks appear across the figures: onRestart() in Figures 4 and 6, and onDestroy() in Figures 7, 8 and 10.

Figure 2. Logcat after launching the app from the emulator.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:21.512 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onCreate()
10-04 05:31:21.558 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStart()
10-04 05:31:21.566 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onResume()
```

Figure 3. Logcat after pressing Home.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:25.912 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onPause()
10-04 05:31:27.082 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStop()
```

Figure 4. Logcat after reopening the app.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:29.263 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onRestart()
10-04 05:31:29.263 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStart()
10-04 05:31:29.264 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onResume()
```

Figure 5. Logcat after switching to Settings.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:32.937 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onPause()
10-04 05:31:33.589 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStop()
```

Figure 6. Logcat after switching back to the app.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:36.650 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onRestart()
10-04 05:31:36.650 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStart()
10-04 05:31:36.651 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onResume()
```

Figure 7. Logcat after rotating to landscape.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:32:17.338 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onPause()
10-04 05:32:17.343 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStop()
10-04 05:32:17.344 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onDestroy()
10-04 05:32:17.615 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onCreate()
10-04 05:32:17.618 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStart()
10-04 05:32:17.623 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onResume()
```

Figure 8. Logcat after pressing Back.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:48.780 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onPause()
10-04 05:31:49.752 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStop()
10-04 05:31:49.754 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onDestroy()
```

Figure 9. Logcat after launching the app again.

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:31:52.360 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onCreate()
10-04 05:31:52.368 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStart()
10-04 05:31:52.370 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onResume()
```

Figure 10. Logcat after tapping Finish Activity, which calls finish().

```
Logcat  emulator-5554 (Pixel 10 Pro) · filter: tag:PamesaLifecycle
10-04 05:32:00.428 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onPause()
10-04 05:32:01.509 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onStop()
10-04 05:32:01.510 6229 6229 D PamesaLifecycle: Dawn Andrei Pamesa is calling onDestroy()
```

Conclusion

The Activity lifecycle follows a fixed order. Opening an Activity always runs `onCreate()`, `onStart()` and `onResume()`, and leaving it always runs `onPause()` before `onStop()`. Whether the Activity comes back through `onRestart()` or `onCreate()` depends on whether it was only stopped or fully destroyed: Home and switching apps only stop it, while Back, `finish()` and rotation destroy it. Rotation was the most surprising result, because the same screen is destroyed and rebuilt, which means any data not saved would be lost. Since Android does

not guarantee `onDestroy()` when it kills the process, important data should be saved in `onPause()` or `onStop()` instead of waiting for `onDestroy()`.